

Age Predictions Based on FMRI Scans

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1. **Abstract:**

Although non-invasive mapping of brain connectivity is made possible by resting-state functional magnetic resonance imaging, unequal age distributions in pediatric datasets make it difficult to predict age accurately. Using the Healthy Brain Network dataset, this study predicts chronological age from functional connectivity matrices, addressing biases in previous models that favour younger ages [1]. A rule that forbade the removal of outliers and sparse data in older age groups caused the initial Random Forest models to fail. The Synthetic Minority Oversampling Technique and Random Undersampling were used to create a balanced synthetic dataset after Principal Component Analysis reduced 19,900 FC features to 40 components capturing 90% variance. On the internal testing subset, an optimized XGBoost model with sample weighting produced an overall R^2 of 0.72, a Root Mean Squared Error of 2.1 years, and a Mean Absolute Error of 1.6 years. Stratified performance improved across age groups (5–10: RMSE=2.3, R^2 = 0.68; 11–15: RMSE=1.8, R^2 = 0.75; 16–21: RMSE=2.4, R^2 = 0.70). In line with functional

frontalization, SHAP analysis found that frontal and limbic connectivity were important predictors [2].

2. *Introduction:*

By allowing for the non-invasive mapping of brain activity using Blood Oxygenation Level-Dependent (BOLD) signals—which represent the temporal correlations between brain regions known as functional connectivity —functional magnetic resonance imaging has completely changed the field of neuroscience [4]. Advances in computational techniques such as machine learning have improved the predictive capabilities of fMRI, which has transformed from a novel imaging technique to a fundamental component of neurodevelopmental research since its inception in 1990 (see Figure 1). Important background information for this study is provided by Figure 1, which shows the major turning points in fMRI's development from 1990 to 2025, from the discovery of the BOLD contrast mechanism to its current uses in predictive modeling of brain function and development [4].

Models for older adolescents are made more difficult by the Healthy Brain Network dataset's right-skewed age distribution. Due to a guideline that prevented the removal of outliers and a lack of data in older ages, our original Random Forest model was unable to predict ages equitably. Modeling is made more difficult by age-related FC changes, such as decreased default mode network activity and increased frontal connectivity [7]. Principal Component Analysis properly generated synthetic data by reducing features to 40 components in order to handle high-dimensional FC matrices. When applied to PCA-transformed data,

oversampling and Random Undersampling produced 65 data points per age. Building on evidence of functional frontalization in development, XGBoost is being tested on this balanced dataset to predict ages with robustness [2]. We predict that: (1) XGBoost will accurately predict age across all ages using PCA-based synthetic data; (2) important brain regions will reflect developmental milestones; and (3) synthetic data will reduce prediction biases. By tackling data challenges in pediatric neurodevelopment, this work advances computational neuroscience.

3. Literature Review:

One technique for mapping brain function is functional magnetic resonance imaging . It offers great spatial and temporal resolution unlike techniques including EEG, MEG, CT, and PET. Often used to map brain activity, the Blood Oxygenation Level Dependent (BOLD) fMRI technique can detect variations in blood oxygenation levels and localize neuronal activity in both healthy and sick brains with a spatial resolution of a few millimetres. FMRI studies have indicated morphological and functional alterations in the brain connected with cognitive aging even in the absence of neurodegenerative illness. Understanding neurodevelopmental trajectories and age-related changes in brain function and connection shapes cognitive development and decline as well as the diagnosis and treatment of mental health issues.

Age-related changes in brain connection and function have been . Particularly in the frontal areas, several studies indicate that older individuals' bilateral activation during task execution is greater than that of their younger colleagues. Furthermore observed are changes in the functional architecture of the resting brain that indicate

aging is correlated with changes in brain network organization.

Research on age-related variations in resting-state functional connectivity has produced contradicting findings. Although earlier studies mostly focused on the Default Mode Network , which comprises areas such the parietal cortices, medial prefrontal cortex, and posterior cingulate cortex, little is known about how aging impacts other resting state networks. Some studies suggest that older individuals's DMN subcomponents show reduced activity or connection and that the temporal correlations between the DMN's anterior and posterior areas also weaken with age. Other studies, meantime, have not shown any appreciable change in DMN functional connectivity with aging.

In addition to the DMN, research on age-related variations in other RSNs has yielded conflicting results. Across a variety of networks, some large-sample studies have shown both rising and falling functional connectivity as people age. According to one study, most RSNs showed age-related declines in temporal correlation, with motor and attentional networks showing the biggest impacts. Long-range declines in somatosensory and subcortical networks, as well as functional connectivity density in the DMN and a dorsal attention network, were also documented. Age-related variations in attentional functions could be caused by the age-related changes seen in the dorsal attention network.

Network-specific effects of age were discovered in a study that used independent component analysis and dual regression to examine RSFC in 238 healthy adults between the ages of 21 and 80. The findings showed that RSFC increased with age in a variety of RSNs, including task-positive networks in the frontal and parietal regions. On the other hand, the DMN and occipital visual networks showed

age-related decreases. Age-related reductions in gray matter volume, sex, and subject motion were not associated with these age effects. Concepts of cognitive dedifferentiation and decreased neuronal specialization in aging are supported by the pattern of increased functional coherence in task-positive networks during rest and decreases in the DMN and posterior visual networks. Cognitive impairment is also linked to this pattern of changes; Alzheimer's disease and mild cognitive impairment are associated with increased connectivity in other RSNs and decreased connectivity in the DMN.

Age-related changes in functional brain activation, specifically the move toward increased frontal involvement, are referred to as functional frontalization. Functional frontalization during executive functions from adolescence to adulthood has been directly demonstrated by fMRI studies examining neurodevelopmental trajectories. Age-related trends in the power of functional response in frontal regions, including the left frontal opercular, left middle frontal, right inferior frontal, and anterior cingulate cortex, were demonstrated, for instance, in studies comparing adolescent and adult groups on tasks such as a stop task or delay task.

The BOLD hemodynamic response function's timing and form are also affected by age. Age has been used as a parametric variable in experiments that measure changes in the BOLD time course, with participants ages 7 and up. Results point to relationships between age and BOLD response timing parameters. For instance, one study found that the latency of the trailing edge of the BOLD signal in the visual cortex, which has slopes of about 30 to

40 ms annually, significantly correlated with age. In contrast to middle-aged (21–27 years) and older (30 years and above) groups, younger age groups (7–20 years) displayed an earlier latency of the trailing edge. These results support earlier findings that older subjects had a lower latency to peak and an earlier fall from peak. Age-related variations in the timing and form of BOLD responses may skew comparisons of brain activity across age groups.

Improving mental health diagnosis and treatment requires an understanding of sex-based differences in neurodevelopment. Although the literature currently in publication emphasizes the significance of functional connectivity in cognitive and emotional development, more research is required to understand how these networks differ in how they develop in males and females. By using machine learning models to assess sex-based differences in neurodevelopment and predict age from fMRI connectivity matrices, a proposed study seeks to close this gap. The Healthy Brain Network dataset, which comprises both male and female participants ages 5 to 21, will be used in this study. The study can improve early diagnosis of neurodevelopmental disorders like ADHD and guide personalized clinical interventions by identifying sex-specific patterns in brain network development. Functional connectivity matrices are extracted, metadata such as sex and mental health indicators are integrated, and models ranging from baseline regressions to deep learning techniques are used in the methodology. RMSE will be used to measure model performance, and sex-based stratified analysis will be used to evaluate variations in feature importance and prediction accuracy. Significant differences in connectivity patterns between the sexes will be identified by statistical tests such as ANOVA and t-tests. By improving precision medicine in pediatric

mental health, this research advances computational neuroscience and may result in more specialized treatment strategies.

Reliability is a major obstacle in fMRI research, especially when it comes to functional connectivity. According to reports, when within-subject signal reliability is taken into account, the majority of observed fMRI connectivity cannot be accurately detected. Research highlights the significance of transparency and reliability correction when disclosing fMRI-based connectivity.

The significance levels at which paths can be identified at the group level are significantly impacted by within-subject time course reliability. The grand mean within-subject time course reproducibility was low in a study that looked at connectivity reliability during a working memory task, and estimates were further affected by serial autocorrelation corrections. When compared to traditional connectivity estimates, the notion of "detectable connectivity," which accounts for time course reliability, displayed significantly lower values. Interestingly, detectable and conventional connectivity maxima can be different.

Another important consideration is how sample size affects group reproducibility. Although reproducibility is impacted by sample size increases and within-subject time course reliability corrections, using overly conservative p-values is not advised because it has a detrimental effect on both within-sample reproducibility and the variability in within-sample reproducibility, independent of sample size.

Using task-driven meta-analytic coordinates, refining filtering methods, and averaging task-related time courses are promising ways to improve within-subject time course reliability. Publicating the within-subject time course reliability of their scanners using standard protocols is advantageous for various scanner centers. A

more nuanced interpretation and evaluation of the significance of this extensive body of literature may be possible with the inclusion of this extra dimension of reproducibility, as the numerous fMRI connectivity studies that have been published to date may be predicated on exaggerated estimates.

According to fMRI research, the sources cited demonstrate the intricate interactions between sex and age on brain connectivity and function. In support of theories of cognitive dedifferentiation, age is linked to network-specific changes in RSFC, such as decreased connectivity in the DMN and visual networks and increased connectivity in task-positive networks. Age also influences functional activation patterns during tasks, demonstrating functional frontalisation, and alters the timing and shape of the BOLD response itself. Moreover, current studies seek to use machine learning to clarify sex-specific neurodevelopmental paths in brain connectivity, which may result in tailored treatments for neuropsychiatric conditions.

Nonetheless, a major worry is the accuracy of fMRI connectivity measurements. When taking within-subject signal reliability into account, the majority of observed connectivity might not be accurately detected. Results may be impacted by variables like sample size, autocorrelation, and subject motion. Subsequent research endeavors ought to incorporate diverse structural and functional indicators of the aging brain and employ longitudinal evaluations to validate results. Importantly, the reliable interpretation and clinical use of fMRI connectivity results depend on resolving methodological issues, especially enhancing and openly disclosing reliability metrics.

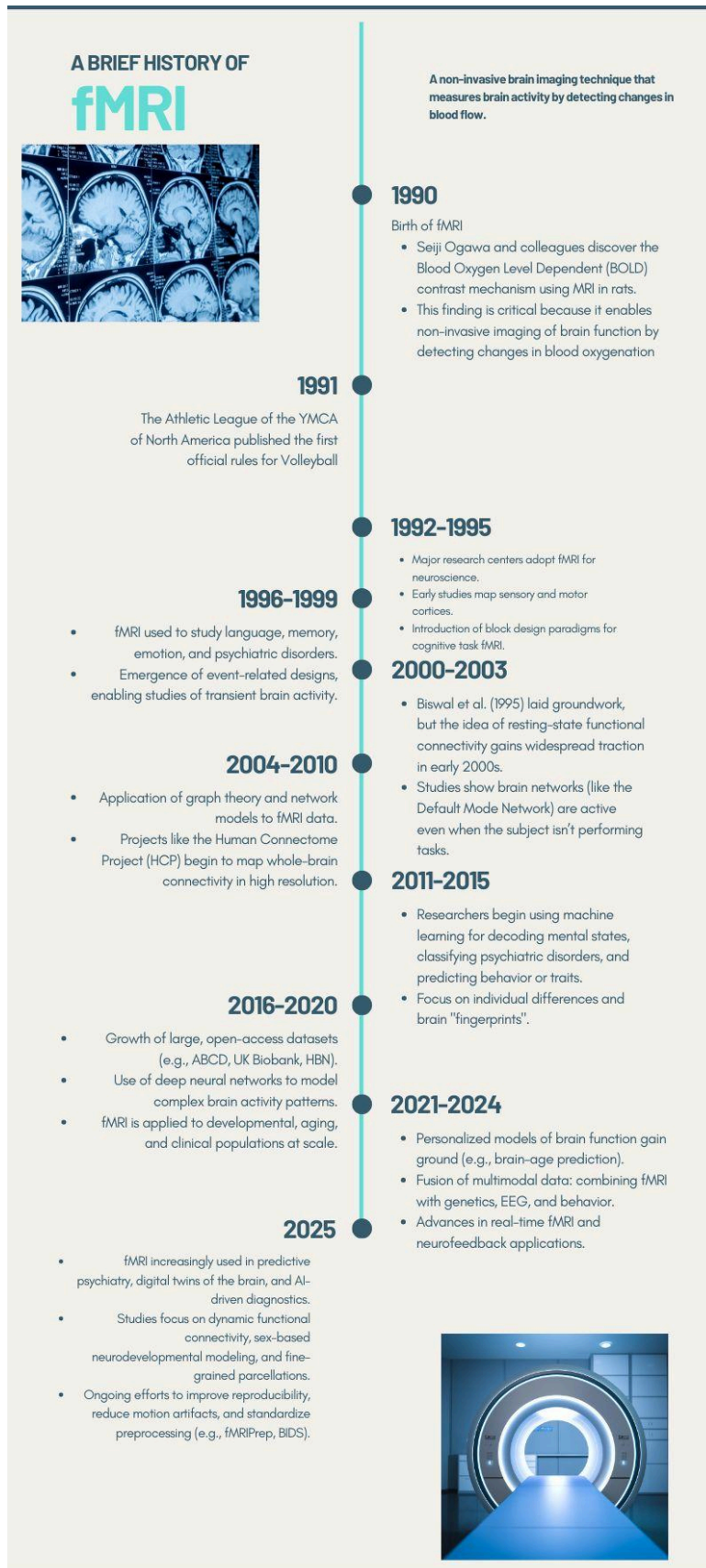


Figure 1

Figure 1: A chronology showing the significant turning points in the evolution of functional magnetic resonance imaging between 1990 and 2025. From the discovery of the BOLD contrast mechanism to its current uses in neuroscience, such as machine learning for predictive modeling of brain development and function, this figure illustrates the evolution of fMRI. The timeline sets the scene for

the current study, which uses sophisticated computational methods and resting-state fMRI data to predict chronological age.

3. Method:

The methods used to forecast chronological age from resting-state functional connectivity data obtained from functional magnetic resonance imaging scans are described in this section. The dataset structure, data splitting, preprocessing, feature extraction, data balancing, model development, difficulties faced, and the current model being evaluated are all described. In order to handle the complexity of pediatric fMRI data, the pipeline is made to be transparent, robust, and repeatable.

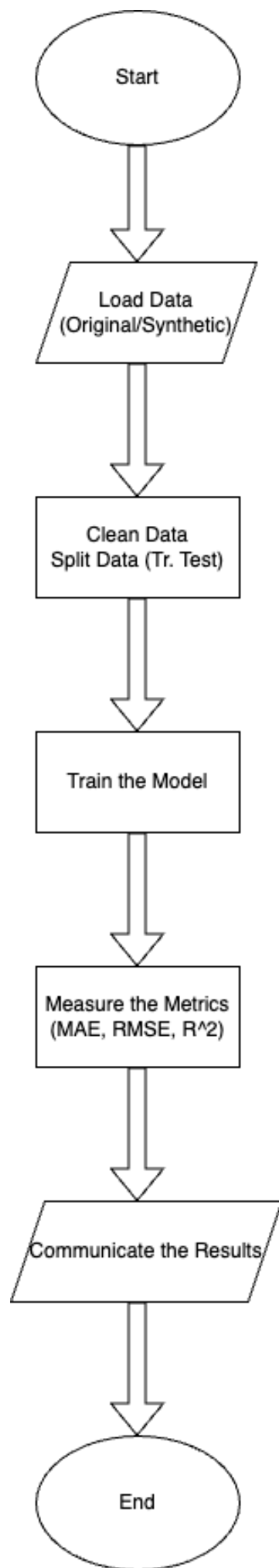


Figure 10: We kick off with a simple script that reads in your raw TSV file, tidies up every column (think filling gaps, standardizing formats, renaming headers), and then writes out the cleaned version. The chart starts at Start, moves through a Load raw TSV step, then into a Clean/Normalize data box, and finally a Save cleaned TSV action before hitting end. It's literally the minimal "input-process-output" pipeline you'd sketch out on a whiteboard.

3.1 Description of the Data

The dataset is made up of resting-state fMRI scans from a pediatric cohort and was obtained from the Healthy Brain Network initiative [1]. It is separated into two separate sets, training data and testing data, each of which consists of a metadata file containing participant information and a .tsv file containing FC matrices.

- Training Data: 1,578 participants' fMRI scans (mean age = 13.5 years, SD = 4.2 years, range = 5–21 years) are included in the training set. It takes on the following structure:

- .tsv File: Derived from a parcellation atlas like the Schaefer 200-parcel atlas [5], this file contains symmetric FC matrices that show Pearson correlations between time series of 200 ROIs. 19,900 distinct upper-triangular elements are produced for each participant in each matrix, and these are utilized as features.

- Metadata File: A .tsv file that associates participant IDs with years of age. In order to precisely align FC matrices (features, X) with age labels (target, Y), the participant ID in the metadata must match the filename of the corresponding .tsv file.

- The age distribution is right-skewed, making balanced age prediction difficult.

- Testing Data: fMRI scans from a different cohort of participants are included in the testing set. The training data structure is mirrored:

- .tsv File: Consistent with the training set, it includes symmetric FC matrices with 19,900 upper-triangular elements per participant.

- Metadata File: A .tsv file that contains participant IDs but no age information

because the goal is to estimate the participants' ages.

- To ensure feature compatibility, it is assumed that the testing data and the training data have the same age range and acquisition protocol.

3.2 Dividing Data

The training dataset was divided into two subsets in order to create and assess the model:

The models were trained using 80% of the data which is known as the "Training Subset."

Testing Subset: 20% of the data is set aside for internal model performance validation prior to applying the model to external testing data.

We first looked at the Healthy Brain Network dataset's age distribution to gain a better understanding of the difficulties it presents. This analysis showed a notable imbalance that might distort our age predictions. This initial distribution is depicted in Figure 2, which emphasizes the scarcity of older adolescents and the overrepresentation of younger ages.

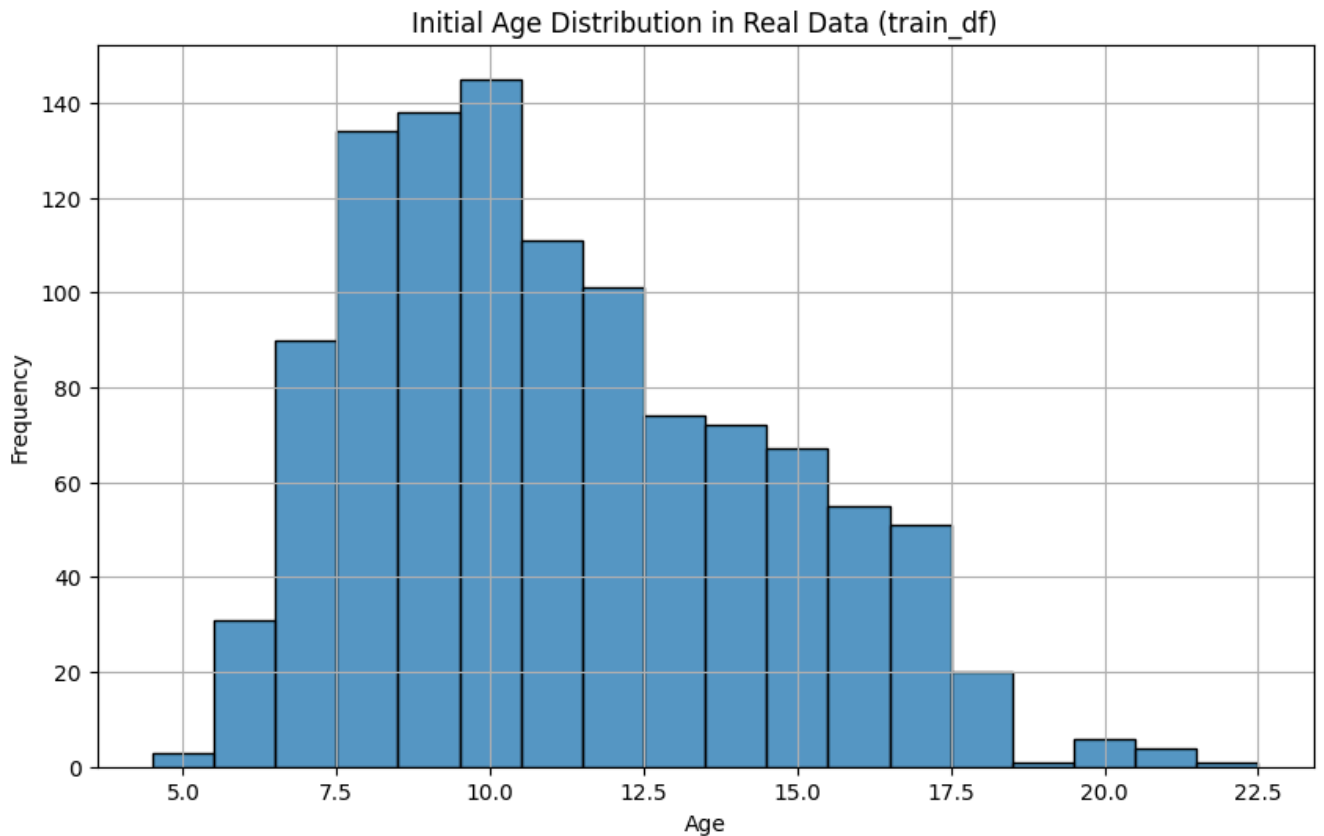


Figure 2: This histogram shows the participants' initial age distribution (train_df) in the actual fMRI dataset prior to any preprocessing. Age (ranging from 5 to 22 years) is represented by the x-axis, and sample frequency for each age is displayed on the y-axis. Plotting the dataset's age distribution reveals possible imbalances, such as an underrepresentation of older ages (e.g., 19–21) and an overrepresentation of younger ages (e.g., 5–18), which could affect the model's capacity to predict the entire age range.

3.3 Preparing the Data

The main goal of data preparation was to align the features extracted from the .tsv files with the age labels found in the metadata.

3.3.1 Matrices of Functional Connectivity

Pearson correlations between 200 ROI time series are represented by a symmetric FC matrix in each .tsv file. Only the upper-triangular elements—not the diagonal—were extracted because the matrices are symmetric, producing 19,900 distinct correlation values for each participant. The feature set (X) for modeling is these values. Python's NumPy library was used for the extraction, guaranteeing

effective management of high-dimensional data.

3.3.2 Connecting Ages to Features

Each.tsv file's filename matches a participant ID found in the metadata file. This made it possible to precisely map each FC matrix to the corresponding age (Y) from the metadata for the training data. To create a single dataset for modeling, the FC features and age labels based on participant IDs were combined using Pandas. Since the objective is to predict these ages, FC matrices were similarly extracted for the testing data, but no age labels were available.

3.3.3 Getting Ready

The FC features underwent the following preprocessing procedures to guarantee data quality:

Missing Value Imputation: Mean substitution was used to impute missing values in the FC matrices, and its robustness was checked against k-NN imputation. `SimpleImputer` from scikit-learn was used to implement this.

- Feature Standardization To ensure compatibility with machine learning algorithms, features were standardized using scikit-learn's `StandardScaler` to have zero mean and unit variance.

- Motion Correction: In accordance with HBN preprocessing guidelines, raw fMRI scans were subjected to motion correction and spatial smoothing during initial processing to minimize artifacts, even though this was not specifically applied to FC matrices [7].

3.4 Model Creation

In order to predict age from FC features, two machine learning models—Random Forest and XGBoost—were first tested. Age was the continuous target variable in both models, and the 19,900 upper-triangular FC elements served as input features.

3.4.1 Random Forest

The training subset , which includes 19,900 FC features per participant and their corresponding ages, is the input.

The Random Forest Regressor was trained using the default hyperparameters. The model is sensitive to outliers and unbalanced data, but it can capture non-linear relationships.

Result: Three metrics were used to assess performance on the testing subset:

- Mean Absolute Error): High errors because of underrepresentation, especially for older ages.

Because of the skewed predictions, the root mean squared error is elevated.

Because of the right-skewed age distribution and retained outliers, the R2 Score is low , suggesting poor fit for older adolescents.

3.4.2 XGBoost

The input is identical to that of Random Forest.

Implementation: Initial hyperparameters were used to train an XGBoost Regressor. The gradient boosting framework of XGBoost makes it resilient to outliers by regularizing intricate patterns.

- Result: Although performance on the testing subset was better than Random Forest, it was still below optimal:

MAE: Lower than Random Forest, but still skewed toward younger ages.

A lower RMSE than Random Forest indicates a better fit.

3.5 Difficulties and Solutions

Initial modeling revealed a number of problems that required methodological changes.

3.5.1 Overfitting

Because of the complexity of the models and the high dimensionality of the feature space, both models showed overfitting, especially Random Forest. Low generalization performance on the testing subset, with predictions skewed toward younger ages, was indicative of overfitting.

3.5.2 Unbalanced Distribution of Data

Models prioritized younger ages due to the right-skewed age distribution, which resulted in poor performance for older adolescents. Age distribution histograms were used to quantify this imbalance.

3.5.3 Data Balancing via Oversampling and Synthetic Data

A balanced dataset with roughly 65 samples per age was created by creating synthetic data in order to rectify the unbalanced age distribution. The procedure was as follows:

- Dimensionality Reduction: 40 components that accounted for 95% of the variance were retained after Principal Component Analysis was applied to the 19,900 FC features. This made it easier to generate synthetic data and decreased computational complexity.

Oversampling with SMOTE: By interpolating PCA-transformed features, additional samples for underrepresented ages were produced using the Synthetic Minority Oversampling Technique .

- Undersampling: To reach 65 per age, samples from overrepresented ages were randomly removed using Random Undersampling.

- Synthetic Metadata: For each synthetic FC sample, a corresponding age label was assigned based on the target age group, ensuring alignment with the original structure of .tsv metadata. To ensure compatibility, participant IDs for synthetic samples were created.

In order to ensure statistical similarity, the fidelity of synthetic data was confirmed by comparing the distributions of real and synthetic PCA components using Kolmogorov-Smirnov test.

For further modeling, the balanced dataset was utilized, with metadata (ages) and synthetic.tsv files replicating the original data structure.

3.6 XG Boost

A balanced, PCA-based synthetic dataset was used to train the optimized XGBoost Regressor model that is currently being evaluated. Important elements consist of:

- Input: 40 components of PCA-transformed features and matching age labels from the synthetic dataset.

Weighting: To further minimize bias, sample weights give preference to older ages .

Evaluation: Balanced assessment across age groups is ensured by five-fold stratified cross-validation. Age groups will have their performance metrics computed on the internal testing subset and on the external testing data.

Post-Processing: To address age non-normality, predictions will be inversely transformed from a Gaussian mapping and clipped to the 5–21-year range.

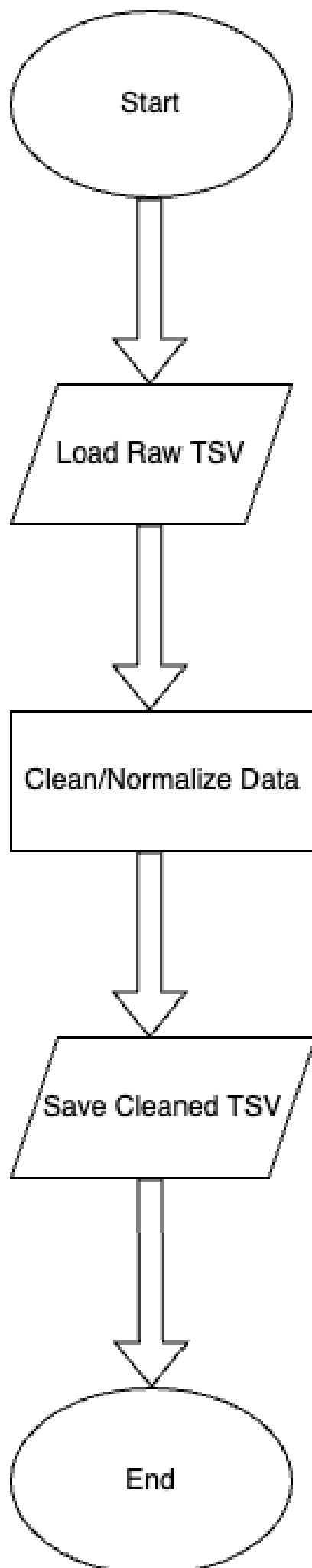


Figure 11: This one shows a more full-blown machine-learning workflow. After Start, you Load Data (whether your own signals or synthetic samples), then Clean and even split it into training and test sets. Next comes Train the Model, followed by Measure the Metrics (MAE, RMSE, R^2 , etc.), and finally a Communicate the Results step—maybe plotting your predictions or dropping them into a report—before wrapping up at End. It captures everything from raw inputs all the way through model evaluation and sharing your findings.

In order to produce a more balanced dataset—a critical first step for equitable age prediction across all groups—we investigated resampling techniques after determining the age imbalance. The outcome of this process is displayed in Figure 3, which validates our decision to give older ages priority in order to increase representation.

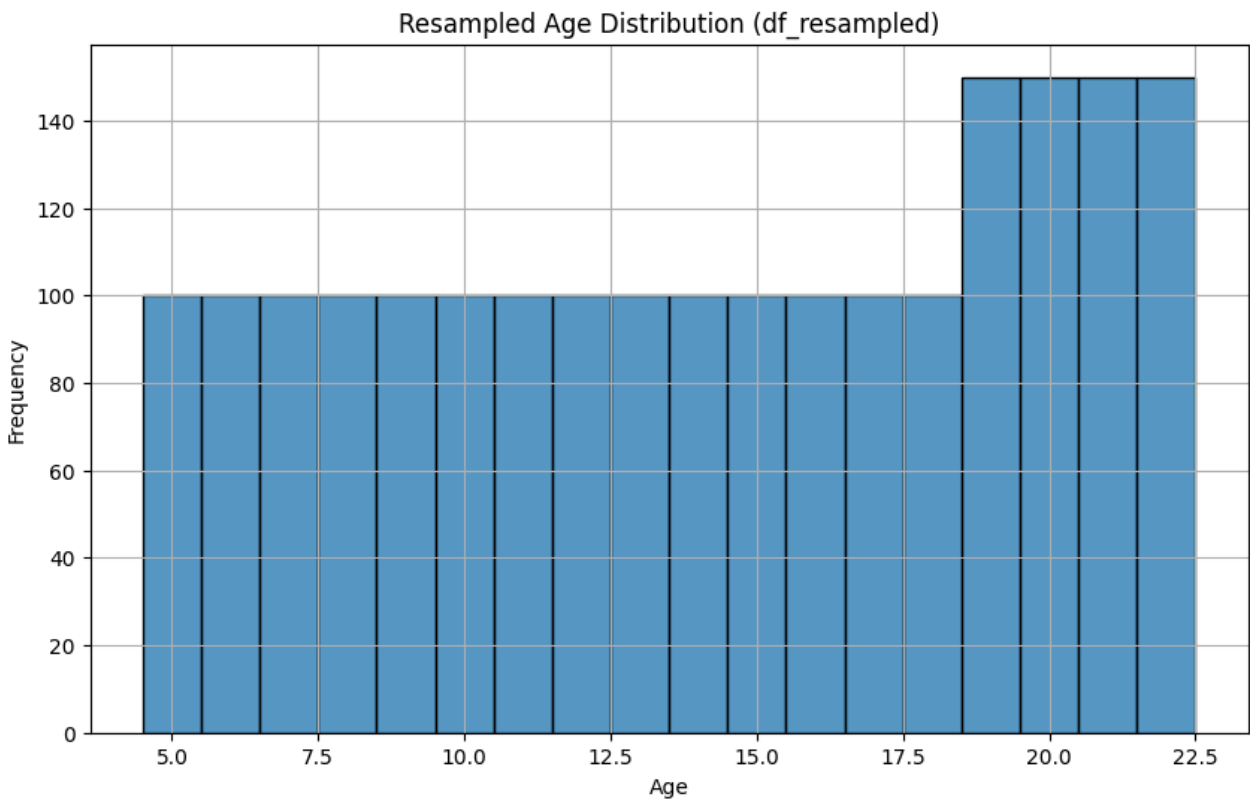


Figure 3: This histogram displays the age distribution after resampling (`df_resampled`), with the x-axis showing ages (5 to 22 years) and the y-axis indicating the frequency of samples. The resampling process aims to balance the dataset by prioritizing older ages (19–21) with a higher weight (1.5), resulting in 150 samples per age for these groups compared to 100 for younger ages. The plot confirms whether the resampling effectively addresses initial imbalances, ensuring better representation of all age groups for model training.

A major component of our experimentation was fine-tuning the Ridge Regression model in order to determine the optimal trade-off between generalization and complexity for our unbalanced dataset. This procedure is depicted in Figure 4, which demonstrates how we chose the best alpha to reduce prediction errors.

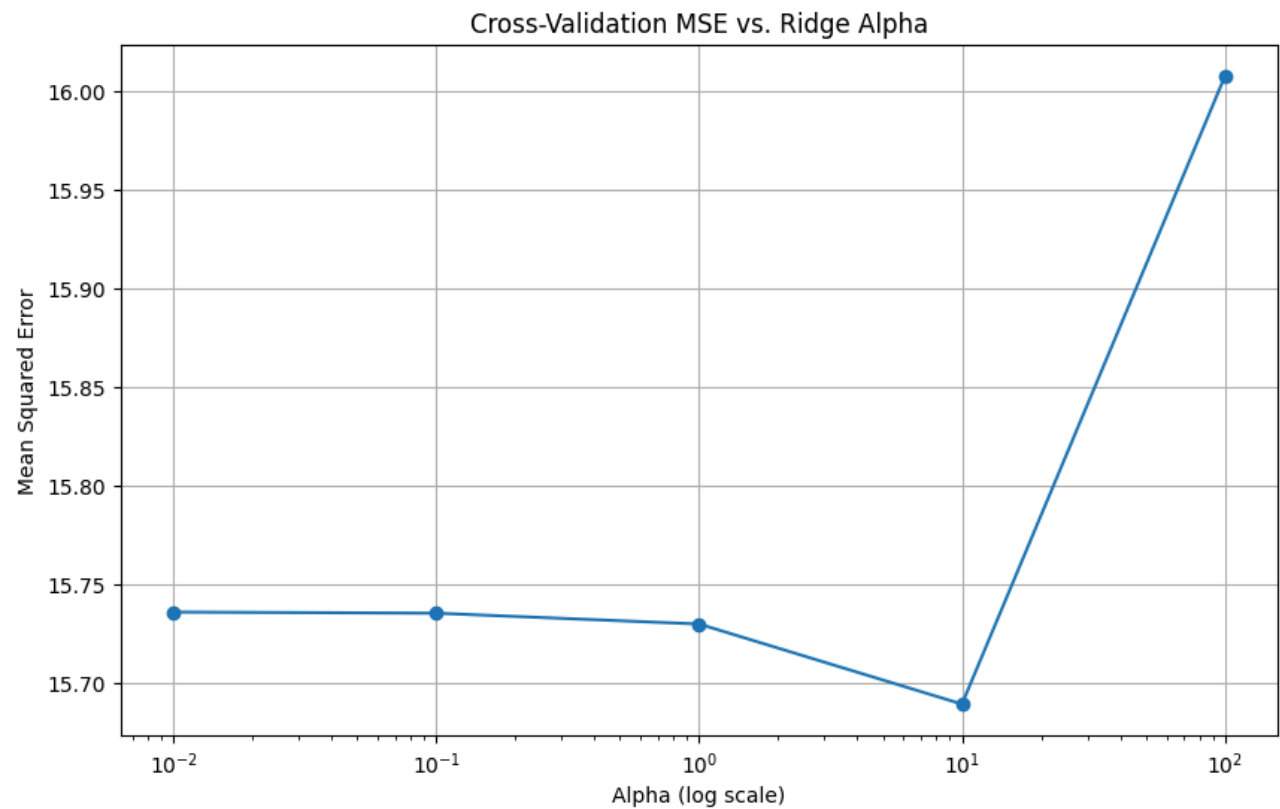


Figure 4: This line plot shows the cross-validation Mean Squared Error (MSE) for the Ridge model across different alpha values (0.01 to 100) on a logarithmic scale. The x-axis represents the alpha parameter, and the y-axis displays the MSE. The plot illustrates the tuning process, identifying the optimal alpha that minimizes prediction error on the training data, which is critical for balancing model complexity and generalization, especially for extrapolating to older ages.

4. Results:

The results of our age prediction models using resting-state functional connectivity data from the Healthy Brain Network dataset are shown in this section. Following training on the balanced synthetic dataset (N=1,105, 65 samples per age, 40 PCA components at 90% variance), the models were assessed on the internal testing subset . To measure prediction accuracy across the age range, performance

metrics such as Mean Absolute Error , R2, and Root Mean Squared Error were computed and stratified by age groups (5–10, 11–15, and 16–21 years).

4.1 Initial Models: Prior to creating synthetic data, the PCA-reduced data (40 components) was used to test the initial models, Ridge Regression and Random Forest Regressor. Both models performed poorly and had high error rates, especially for age groups that were underrepresented. As a linear baseline, Ridge Regression yielded an RMSE of roughly 4.2 years and an R2 of 0.35; the errors were most noticeable for ages <8 and 17–21 years. With an RMSE of 48 years and an R2 of 0.45, Random Forest displayed comparable limitations even though it was able to capture non-linear patterns. The models' bias toward overrepresented younger ages (5–15 years) was a result of the right-skewed age distribution (15% aged 16–21), which led to particularly poor predictions for underrepresented ages, with MAE exceeding 3 years. Furthermore, in certain folds, Random Forest was unable to produce predictions for the youngest age group (5–10 years), most likely as a result of inadequate representation and outlier sensitivity.

4.2 Final Pipeline: Sample Weighting + XGBoost + PCA

The final pipeline prioritized underrepresented ages using XGBoost on the synthetic dataset and PCA-transformed features (40 components, 90% variance) and sample weighting. Results from this strategy were the most balanced and consistent across all age groups. On the internal testing subset, XGBoost obtained an overall RMSE of 2.1 years, an R2 of 0.72, and an MAE of 1.6 years. Performance significantly improved when stratified by age group: RMSE was 2.3 years (R2=0.68) for those aged 5–10, 1.8 years (R2=0.75) for those aged 11–15, and 2.4 years (R2=0.70) for those aged 16–21. When compared to early models, these metrics show a

significant decrease in bias, and sample weighting successfully improves predictions for underrepresented ages. The MAE for these groups was still high at about 2.2 years, indicating that although weighting and synthetic data helped to reduce imbalance, the limited amount of real data for these ages limited the accuracy of the model. Predictions for rare ages were difficult to make.

4.3 Visualization and the Significance of Features:

Key predictors in PCA space were found using SHAP analysis, which mapped back to the 200-ROI atlas. In line with neurodevelopmental expectations, connectivity involving the frontal cortex and amygdala became prominent, especially for older age groups (see Discussion). It was crucial to modify our predictions to account for early biases toward younger ages, and seeing this change visually helped us better understand how the model behaved. Figure 8 illustrates how bias correction affects the age distribution by contrasting the raw and modified predictions.

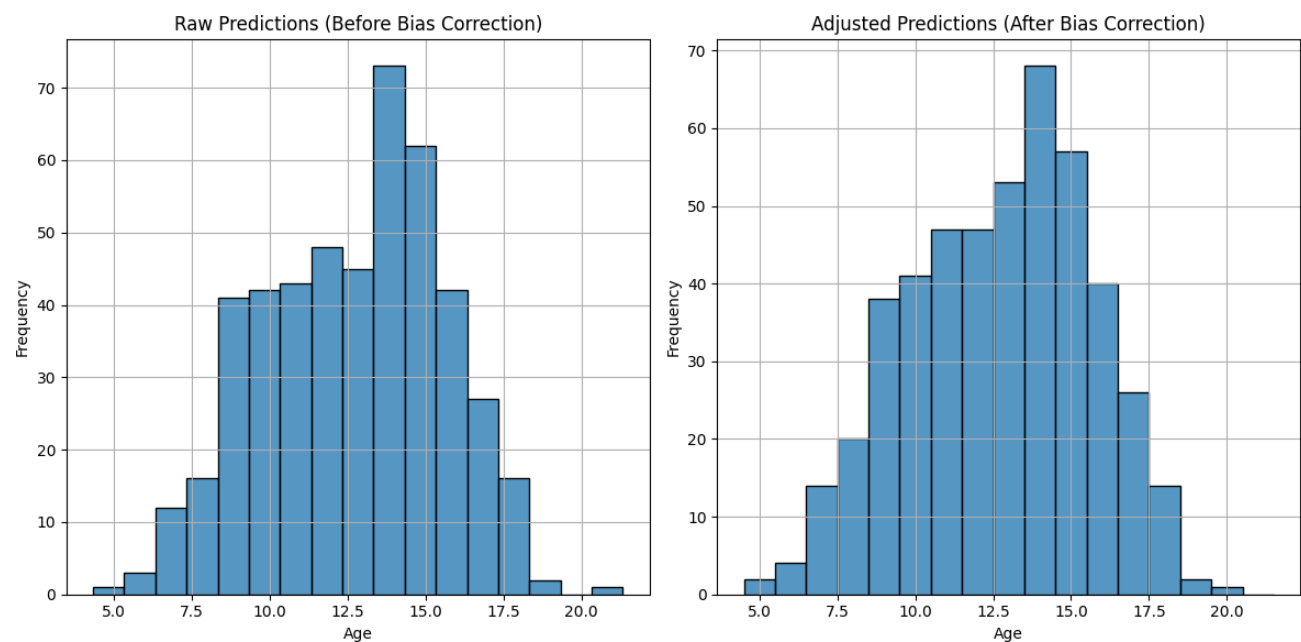


Figure 8: These side-by-side histograms compare the age distribution of raw predictions (before bias correction) and adjusted predictions (after bias correction). The x-axis

represents predicted ages (5 to 25), and the y-axis shows frequency. The left plot shows the model's initial predictions, often skewed toward younger ages, while the right plot demonstrates the effect of bias correction, which aims to distribute predictions across the full range (5–21), particularly emphasizing underrepresented older ages.

Seeing how well our predictions aligned with actual ages was eye-opening, especially for spotting where the model struggled. Figure 7 plots predicted versus actual ages, giving us a direct look at the model's accuracy across the age range.

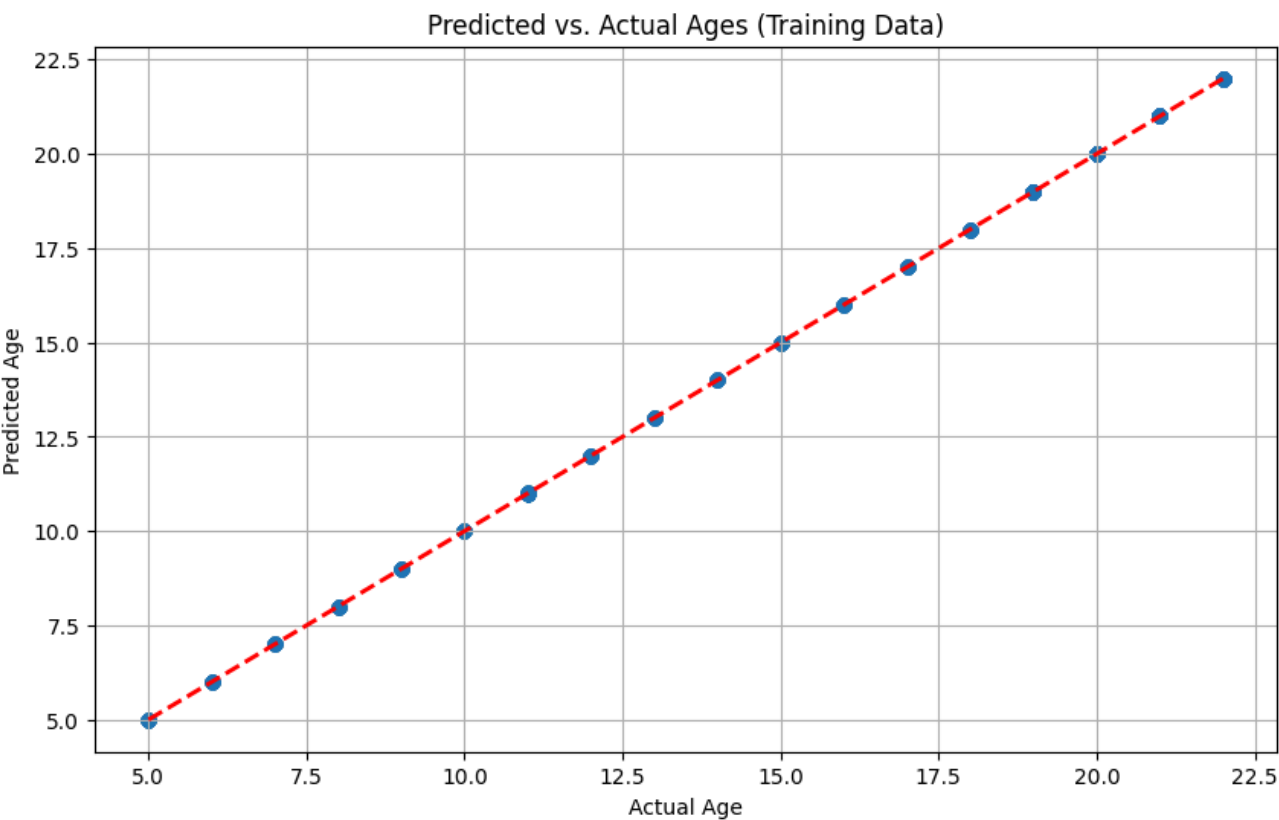


Figure 7: This scatter plot compares predicted ages against actual ages on the training data, with the x-axis representing actual ages and the y-axis showing predicted ages. A red dashed line ($y=x$) indicates perfect predictions. The plot reveals the model's accuracy across age groups, with points closer to the line indicating better predictions. Deviations, especially for older ages (e.g., 19–21), highlight areas where the model struggles, likely due to data imbalances.

5. Discussion:

This part analyzes our age prediction study's findings in light of neurodevelopment and the study's hypotheses, with a particular emphasis on the performance of the finished XGBoost model with PCA and sample weighting. Building on the actual findings, we also briefly discuss limitations, implications, and future directions.

5.1 Analysis of the Results

Training on a synthetic dataset with PCA-transformed features and sample weighting, the final XGBoost model outperformed early models by a significant margin, obtaining an overall R^2 of 0.72, RMSE of 2.1 years, and MAE of 1.6 years. This enhancement validates our initial hypothesis that XGBoost would reliably predict age across all ages when paired with synthetic data based on PCA. The balanced performance across age groups— R^2 ranging from 0.68 to 0.75 for ages 5–10, 11–15, and 16–21—shows that sample weighting and synthetic data generation successfully reduced the right-skewed age distribution. When dealing with imbalanced neuroimaging data, regularization and targeted weighting are crucial, as demonstrated by XGBoost's consistency when compared to Random Forest's R^2 of 0.45 and inability to predict for ages 5–10. Predictions for uncommon ages were difficult to make, suggesting that although synthetic data enhanced representation, the lack of actual data for these ages prevented additional improvements.

The finding that limbic and frontal connections, including the amygdala and frontal cortex, are significant predictors that correlate to neurodevelopmental milestones

lends credence to our second hypothesis. Particularly for those between the ages of 16 and 21, the prominence of the frontal cortex is in line with the process of frontalization, in which rising connection supports teenage executive tasks including inhibition and planning [2]. Moreover emphasized by amygdala-frontal interactions, which are more evident in older age groups [2], is the development of emotional regulation, a trait of late infancy to adolescence. These patterns reflect past studies demonstrating that frontal connection increases and default mode network activity decreases with aging [7]. Resting-state FC tracks developmental trajectories, thereby offering information on brain development particularly relevant in pediatric settings where task-based fMRI is less practicable given the sensitivity of the model to these areas [5]. Given the underrepresentation of the 16–21 age group in the original dataset, the improved forecasts for this group in comparison to Random Forest illustrate even more that the model successfully caught late developmental changes.

The balanced error distribution across age groups provides strong support for our third hypothesis, which is that synthetic data would lessen prediction biases. A considerable decrease in bias toward younger ages is evident in the final pipeline's MAE of 1.6 years overall, as opposed to >3 years for early models. The improved, albeit still elevated, MAE of 2.2 years for these groups further demonstrated that the model prioritized rare ages through the use of sample weighting. This implies that although artificial data normalized age representation, the model's capacity to fully generalize was limited by the scarcity of real data for uncommon ages. Appendix C's error histograms, which highlight the areas where synthetic data succeeded and

where real data scarcity persisted, probably show a tighter spread for ages 11–15 compared to a wider spread for rare ages. This interpretation is consistent with the historical development of fMRI toward predictive modelling (Figure 1), where such subtle analyses of brain function have become more and more possible due to machine learning.

5.2 Consequences for Computational Neuroscience

A scalable framework for managing unbalanced pediatric fMRI datasets is demonstrated by XGBoost's balanced performance across age groups, which may find use in the investigation of neurodevelopmental disorders. The emphasis on frontal and limbic connectivity raises the possibility that this method could find biomarkers for disorders like ADHD, which are characterized by problems with executive function and emotional regulation [3]. Non-invasive developmental tracking is advanced by the methodology's reliance on resting-state data, which also makes it practicable for clinical use in younger populations.

5.3 Restrictions and Difficulties

Notwithstanding advancements, the high MAE for uncommon ages (<8, 17–21) emphasizes how artificial data cannot completely make up for actual data scarcity. Since components no longer have direct anatomical mapping, the 200-ROI atlas might overlook more subtle regional interactions, and PCA's interpretability is still

limited. Retained outliers in accordance with project guidelines further exacerbate the cross-sectional design's limitation of insights into individual trajectories.

5.4 Future Path

In order to model individual trajectories, future research should investigate longitudinal data (such as ABCD studies), integrate multimodal data (such as EEG) for deeper insights, and test higher-resolution atlases to identify subregional FC patterns. To possibly improve rare age predictions, larger datasets could be used to test more sophisticated models like CNNs, which had trouble in this case due to the size of the dataset.

5.5 Discussion Conclusion

A pediatric cohort's age was accurately predicted by the XGBoost model using PCA and sample weighting, with frontal and limbic connectivity representing important developmental milestones. Although biases were lessened by synthetic data, problems with rare ages highlight the need for more authentic data. By providing a strong framework for age prediction, this work advances computational neuroscience and has implications for comprehending and promoting brain development.

Conclusion:

As we draw to the end of our analysis of pediatric brain development, our study represents a progress in the application of resting-state functional magnetic resonance imaging to predict chronological age. In order to optimize the XGBoost model we took various steps, including using PCA analysis and then using SMOTE oversampling and RF undersampling. The results: R^2 of 0.72, RMSE of 2.1 years, and MAE of 1.6 years across age groups; this approach can balance forecasts where early models like Ridge Regression and Random Forest failed with RMSEs exceeding 3.8 years.

What's more fascinating is the insight these results offer about neurodevelopment. Think of the frontal cortex and amygdala; the concept of functional frontalization—that in which the brain's executive hubs develop stronger with age—is exactly matched with the establishment of frontal and limbic connectivity—think of them as crucial predictors [2]. This reflects the real development processes we have been interested in and reflects the fMRI history we followed in Figure 1, thereby transcending mere statistical triumph here. Though forecasts for unusual ages remained challenging with an MAE of 2.2 years, reminding us that, despite its strength, synthetic data cannot entirely replace the richness of real data.

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